

NEXUS Prime-Gap Program: Canonical Closure Ledger

Integrated Final State Through Phase 10 + Extension

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Abstract

This document closes the **current program state** of the NEXUS prime-gap branch by merging the corrected Phase 10 writeup with the Phase 10 Extension into one canonical ledger.

The purpose of this document is precision. It does **not** claim that the deep classical frontier is solved. It does claim that the current NEXUS branch is now cleanly separated into:

1. a **theorem-grade algebraic core**,
2. a **correctly keyed empirical shell**,
3. a set of **identified analytic mechanisms**,
4. a sharply delimited **remaining frontier**.

The main result is that the branch is no longer a stack of overlapping claims. The present closure state is:

- the **Primorial Family Lattice Theorem** is locked,
- the **Step Theorem** is locked,
- the **exact subtype-count law** is locked,
- the $k = 2$ **renewal shell** is corrected and stable,
- the **spike signs** are analytically positive,
- the $\pi_{30}(X)$ **deficit law** is numerically locked and mechanistically decomposed,
- the **body correction** has a concrete wheel-overflow mechanism,
- the $k = 30$ **branch** now has a body/tail-aware replacement shell,
- the **subtype Hardy–Littlewood asymptotic** is stated precisely,
- the remaining open problems are now specific and finite.

This is the canonical closure of what the project has so far.

1. Closure Convention

In this document, **closure** means one of four things:

1.1 Locked

A statement that is theorem-grade or exact within the branch.

1.2 Fitted

A statement that is numerically stable and internally consistent, but still empirical.

1.3 Mechanized

A statement for which the missing mechanism has been identified, even if the final derivation is incomplete.

1.4 Open

A statement that still requires new analytic work.

The purpose of this ledger is to prevent theorem, fit, mechanism, and speculation from bleeding into one another.

2. Locked Algebraic Core

Let W be a primorial wheel and define

$$U_W = (\mathbb{Z}/W\mathbb{Z})^\times.$$

For an even gap k , define the admissible subtype set

$$S_W(k) = \{ r \in U_W : r + k \in U_W \pmod{W} \}.$$

For each prime pair $(p, p + k)$ in subtype r , define the midpoint center

$$H = p + \frac{k}{2}.$$

Then the theorem-grade kernel is:

2.1 Family Lattice Theorem

$$H \equiv r + \frac{k}{2} \pmod{W}.$$

2.2 Step Theorem

For consecutive midpoint centers in a fixed subtype,

$$\Delta H \equiv 0 \pmod{W}.$$

2.3 Exact subtype count

$$|S_W(k)| = \prod_{\substack{q|W \\ q>2 \\ q \nmid k}} (q-2) \prod_{\substack{q|W \\ q>2 \\ q|k}} (q-1).$$

These three statements are the algebraic floor of the branch.

2.4 Closure state

LOCKED

3. Correct Variable Definitions

The later corrective phases showed that most instability in the branch was not caused by bad algebra. It was caused by bad variable identification.

3.1 Intra-subtype midpoint-gap variable

The correct renewal variable is

$$M = \frac{\Delta H}{W},$$

where ΔH is the midpoint difference between **consecutive centers in the same subtype**.

Equivalently, if

$$H_1^{(r)} < H_2^{(r)} < H_3^{(r)} < \dots$$

are the ordered midpoint centers for subtype r , then

$$M_n^{(r)} = \frac{H_{n+1}^{(r)} - H_n^{(r)}}{W} \in \mathbb{N}_{>0}.$$

This is **not** the same as:

- pooled raw inter-arrivals,
- pooled startpoint gaps divided by W ,
- or any subtype-unaware spacing variable.

3.2 Correct definition of $\pi_{30}(X)$

For the twin-prime family at $W = 30$, with subtype labels

$$r_1, r_2, \dots, r_N \in \{11, 17, 29\},$$

the correct same-subtype adjacency statistic is

$$\pi_{30}(X) = \frac{1}{N-1} \sum_{j=1}^{N-1} \mathbf{1}_{r_j=r_{j+1}}.$$

This is the fraction of **consecutive twin-prime pairs** that land in the same subtype.

It is not the max-class deviation from $1/3$.

3.3 Correct handling of the $k = 30$ branch

The same midpoint-gap construction must be used for $k = 30$:

$$M_{k=30} = \frac{\Delta H}{30}$$

within fixed subtypes.

3.4 Closure state

LOCKED

4. The Corrected $k = 2$ Renewal Shell

The twin-prime family at $W = 30$ is the best-resolved empirical branch.

4.1 NB base kernel

The corrected shell lives on

$$M \in \{1, 2, 3, \dots\}.$$

The base kernel is

$$\text{NB}(m; r, p) = \binom{m+r-2}{m-1} (1-p)^{m-1} p^r, \quad m \geq 1.$$

This corresponds to

$$M - 1 \sim \text{NB}(r, p)$$

in the standard zero-based convention.

Hence the correct moments are

$$\mathbb{E}[M] = 1 + \frac{r(1-p)}{p}, \quad \text{Var}(M) = \frac{r(1-p)}{p^2}.$$

4.2 Additive spike law

The canonical spike law is additive:

$$f_{\text{spike}}(m) = 1 + \alpha_7 \mathbf{1}_{7|m} + \alpha_{11} \mathbf{1}_{11|m} + \alpha_{13} \mathbf{1}_{13|m}.$$

4.3 Empirical body correction

The current working body correction is

$$f_{\text{body}}(m) = 1 + \gamma \mathbf{1}_{6 \leq m \leq 15}.$$

This is an empirical correction, not yet theorem-grade.

4.4 Full corrected shell

The active $k = 2$ shell is

$$\mathbb{P}(M = m) = \frac{1}{Z} \text{NB}(m; r, p) f_{\text{spike}}(m) f_{\text{body}}(m),$$

with fitted values

$$\begin{aligned} r &= 1.021, & p &= 0.050, \\ \alpha_7 &= +0.488, & \alpha_{11} &= +0.208, & \alpha_{13} &= +0.240, \\ \gamma &= +0.075. \end{aligned}$$

4.5 Closure state

FITTED

5. Spike Signs

This is the strongest analytic gain beyond the core algebra.

Let g be the gap between consecutive twin-prime startpoints and let $q > 5$ be prime.

5.1 Generic case: $q \nmid g$

The forbidden residue set

$$\{0, 2, g, g + 2\} \pmod{q}$$

contains four distinct classes, so the local factor is proportional to

$$\frac{q-4}{q}.$$

5.2 Spike case: $q \mid g$

Then 0 and g coincide mod q , and 2 and $g+2$ coincide mod q , leaving only two occupied classes. The factor becomes

$$\frac{q-2}{q}.$$

Thus the enhancement ratio is

$$\frac{(q-2)/q}{(q-4)/q} = \frac{q-2}{q-4}.$$

Since

$$\frac{q-2}{q-4} > 1 \quad \text{for all } q > 5,$$

the spike signs must be positive.

5.3 Consequence

$$\alpha_q > 0 \quad \text{for the relevant } k=2 \text{ spike primes } q > 5.$$

5.4 Closure state

$$\boxed{\text{LOCKED for sign, OPEN for exact finite-}X \text{ amplitude}}$$

6. Equal-Split and Subtype Symmetry

For the $k=2$, $W=30$ family, the admissible subtype residues are

$$\{11,17,29\}.$$

The asymptotic equal-split target is

$$R_\tau(X) \rightarrow 1, \quad \tau \in \{11,17,29\},$$

where $R_\tau(X)$ denotes subtype density relative to perfect equal split.

6.1 Structural statement

The admissible set $S_W(k)$ is acted on by the automorphism structure of the reduced residue system. For $k = 2$, $W = 30$, the subtype set behaves as one orbit under the wheel symmetry, which is the structural reason the equal-split law is the correct asymptotic target.

6.2 Conditional reading

The equal-split law follows from orbit symmetry **once** subtype infinitude / asymptotic existence is granted.

So the clean statement is:

equal-split follows from wheel symmetry given subtype asymptotic existence.

6.3 Closure state

MECHANIZED

7. The $\pi_{30}(X)$ Deficit Law

Define the same-subtype adjacency deficit

$$D_{30}(X) = \frac{1}{3} - \pi_{30}(X).$$

The numerically locked law is

$$D_{30}(X) = \frac{0.104115}{\ln X} + \frac{6.662432}{\ln^2 X}.$$

Equivalently,

$$\frac{1}{3} - \pi_{30}(X) = \frac{0.104115}{\ln X} + \frac{6.662432}{\ln^2 X}.$$

7.1 What is locked numerically

- the statistic,
- the sign,
- the two-term structure,
- consistency with the corrected large-scale runs.

7.2 Mechanism: two-layer decomposition

The extension identifies two analytically distinct layers.

Layer 1 — competing-renewal process structure

For three identical subtype-renewal processes with common inter-arrival PMF $P(M = m)$, the same-component consecutive probability is

$$\pi_{\infty} = \sum_{m=1}^{\infty} P(M = m) [P(R > m)]^2,$$

where the forward recurrence distribution satisfies

$$P(R = r) = \frac{P(M \geq r)}{\mathbb{E}[M]}, \quad r \geq 1.$$

In the geometric approximation this yields

$$\pi_{\infty}^{\text{geo}}(p) = \frac{p(1-p)^2}{1-(1-p)^3},$$

so

$$D_{\infty}^{\text{geo}}(p) = \frac{1}{3} - \pi_{\infty}^{\text{geo}}(p) \approx \frac{p}{3} + O(p^2).$$

This gives an upper-side bracket for the observed A coefficient.

Layer 2 — Hardy–Littlewood singular-series weighting

Same-subtype and cross-subtype transitions carry different 4-tuple singular-series weights. The weighted finite- X approximation is

$$\pi_{30}(X) \approx \frac{\sum_{g \leq G(X)} S(g) \ln^{-2}(X/g)}{\sum_{g \leq G(X)} S(g) \ln^{-2}(X/g)}.$$

This gives the lower-side bracket for the observed A coefficient.

7.3 Current closure state

FITTED: $D_{30}(X)$

MECHANIZED: competing renewal + HL weighting

OPEN: exact analytic derivation of $A = 0.104115$.

8. Body Correction Mechanism

The body term is no longer just a fitted patch. It has a concrete proposed source.

8.1 Wheel-overflow from $q = 7$

The wheel

$$W = 30 = 2 \cdot 3 \cdot 5$$

screens divisibility by 2,3,5 but is silent on the first unscreened prime

$$q_{\text{next}} = 7.$$

This induces a systematic overflow scale

$$m_* \approx \frac{W}{q_{\text{next}}} = \frac{30}{7} \approx 4.3,$$

and a doubled scale

$$2m_* \approx 8.6.$$

So the primary overflow window sits roughly in the lower body, exactly where the residual excess concentrates.

8.2 Empirical meaning

The spike at $m = 7$ captures the exact resonance.

The body term captures the smooth excess around that resonance and its spill into the nearby zone.

8.3 Analytic path

The document's proposed derivation path is to extend the wheel from

$$W = 30 \quad \text{to} \quad W' = 210 = 2 \cdot 3 \cdot 5 \cdot 7,$$

then compare the induced PMFs in the $m \in [6,15]$ region.

8.4 Current closure state

MECHANIZED: body correction as wheel-overflow from $q = 7$

OPEN: first-principles derivation of γ .

9. The $k = 30$ Branch

This branch is no longer summarized by a single-NB story.

9.1 Historical single-NB shell

The old fit is reproducible:

$$r \approx 0.534, \quad p \approx 0.0272.$$

But the mean-excess diagnostic showed this shell was not tail-faithful.

9.2 Two-regime replacement model

The extension introduces a body/tail-aware shell:

$$\mathbb{P}(M = m) = \begin{cases} \frac{w_b \text{NB}(m; r_b, p_b)}{\sum_{j=1}^T \text{NB}(j; r_b, p_b)} & 1 \leq m \leq T, \\ w_t p_t (1 - p_t)^{m-T-1} & m > T. \end{cases}$$

with fitted parameters

$$\begin{aligned} T &= 62, & r_b &= 0.480, & p_b &= 0.021, \\ p_t &= 0.056, & w_b &= 0.972, & w_t &= 0.028. \end{aligned}$$

9.3 Interpretation

- the body is sub-geometric and over-dispersed,
- the tail is much closer to geometric,
- the old single shell collapsed two regimes into one parameter pair.

9.4 Closure state

FITTED: two-regime shell

MECHANIZED: body/tail split

This is the best current shell for $k = 30$.

10. Subtype Hardy–Littlewood Asymptotics

For each admissible subtype $r \in S_W(k)$, define

$$\pi_r(X) = \#\{p \leq X: p \equiv r \pmod{W}, p \text{ and } p+k \text{ prime}\}.$$

The precise subtype asymptotic statement is:

$$\pi_r(X) \sim \frac{1}{|S_W(k)|} C_k \text{Li}_2(X), \quad X \rightarrow \infty.$$

For the twin-prime case at $W = 30$,

$$\pi_r(X) \sim \frac{C_2}{3} \frac{X}{\ln^2 X}, \quad r \in \{11, 17, 29\}.$$

10.1 What is structural

The subtype partition and equal-split factor

$$\frac{1}{|S_W(k)|}$$

are structurally correct.

10.2 What is still classical-hard

The asymptotic rate still depends on the underlying prime-pair existence problem. So this branch ultimately inherits the difficulty of the twin-prime / Polignac frontier.

10.3 Closure state

MECHANIZED: precise subtype HL statement

OPEN: proof of infinitude and full asymptotic rate.

11. Retired Claims

The following are no longer active truth claims in the canonical ledger:

1. pooled raw inter-arrival definitions of M ,
2. max-class deviation as the definition of π_{30} ,
3. Poisson as the base renewal law,
4. Hawkes / excitatory clustering as the explanation of the deficit,
5. shared finite- X kernel for all observed corrections,
6. strong persistent ToA/ToB directional bias,
7. the claim that the old single-NB $k = 30$ shell correctly captures the tail.

These are retired and should not be carried forward into canonical statements.

12. Canonical State Map

The full branch now compresses to:

algebra locked

variables locked

$k = 2$ shell corrected

spike signs positive

π_{30} numerically locked, mechanism identified

body correction mechanism identified

$k = 30$ body/tail shell fitted

subtype HL stated precisely

deep number theory still open

13. Remaining Open Problems

The remaining frontier is now short and precise.

Open Problem 1 — Exact A coefficient

Prove or compute

$$A = 0.104115$$

in the expansion

$$\pi_{30}(X) = \frac{1}{3} + \frac{a}{\ln X} + \frac{b}{\ln^2 X} + O\left(\frac{1}{\ln^3 X}\right), \quad a = -A.$$

Open Problem 2 — Derive γ from the $W' = 210$ lift

Compute the body correction from the explicit wheel extension:

$$W = 30 \rightarrow W' = 210.$$

Open Problem 3 — Analytic origin of the $k = 30$ threshold

Explain the onset threshold T in the two-regime shell from first principles.

Open Problem 4 — Infinitude of each subtype family

For every admissible $r \in S_W(k)$, prove

$$\pi_r(X) \rightarrow \infty \quad \text{as } X \rightarrow \infty.$$

This is the deepest unresolved branch.

14. Final Closure Statement

The canonical closure of the NEXUS prime-gap branch is therefore:

The algebraic scaffold is complete.

The empirical twin-prime shell is coherent.

The main residual corrections now have identified mechanisms.

The remaining open problems are sharply posed analytic tasks, not conceptual confusion.

That is the present closure state.

Appendix A. Core Equations

A.1 Admissible subtype set

$$S_W(k) = \{ r \in U_W : r + k \in U_W \pmod{W} \}.$$

A.2 Family Lattice Theorem

$$H \equiv r + \frac{k}{2} \pmod{W}.$$

A.3 Step Theorem

$$\Delta H \equiv 0 \pmod{W}.$$

A.4 Exact subtype count

$$|S_W(k)| = \prod_{\substack{q|W \\ q>2 \\ q \nmid k}} (q-2) \prod_{\substack{q|W \\ q>2 \\ q|k}} (q-1).$$

A.5 Midpoint-gap variable

$$M = \frac{\Delta H}{W}.$$

A.6 NB base kernel

$$\text{NB}(m; r, p) = \binom{m+r-2}{m-1} (1-p)^{m-1} p^r.$$

A.7 Correct moments

$$\mathbb{E}[M] = 1 + \frac{r(1-p)}{p}, \quad \text{Var}(M) = \frac{r(1-p)}{p^2}.$$

A.8 Additive spike law

$$f_{\text{spike}}(m) = 1 + \alpha_7 \mathbf{1}_{7|m} + \alpha_{11} \mathbf{1}_{11|m} + \alpha_{13} \mathbf{1}_{13|m}.$$

A.9 Body term

$$f_{\text{body}}(m) = 1 + \gamma \mathbf{1}_{6 \leq m \leq 1}.$$

A.10 Full $k = 2$ shell

$$\mathbb{P}(M = m) = \frac{1}{Z} \text{NB}(m; r, p) f_{\text{spike}}(m) f_{\text{body}}(m).$$

A.11 Spike enhancement ratio

$$\text{Enhancement}(q) = \frac{q-2}{q-4}, \quad q > 5.$$

A.12 Same-subtype adjacency statistic

$$\pi_{30}(X) = \frac{1}{N-1} \sum_{j=1}^{N-1} \mathbf{1}_{r_j=r_{j+1}}.$$

A.13 Deficit law

$$\frac{1}{3} - \pi_{30}(X) = \frac{0.104115}{\ln X} + \frac{6.662432}{\ln^2 X}.$$

A.14 Competing-renewal formula

$$\pi_{\infty} = \sum_{m=1}^{\infty} P(M = m) [P(R > m)]^2.$$

A.15 Forward recurrence law

$$P(R = r) = \frac{P(M \geq r)}{\mathbb{E}[M]}, \quad r \geq 1.$$

A.16 Geometric approximation

$$\pi_{\infty}^{\text{geo}}(p) = \frac{p(1-p)^2}{1 - (1-p)^3}.$$

A.17 Weighted HL transition formula

$$\pi_{30}(X) \approx \frac{\sum_{g \leq G(X)} S(g) \ln^{-2}(X/g)}{\sum_{g \leq G(X)} S(g) \ln^{-2}(X/g)}.$$

A.18 Two-regime $k = 30$ shell

$$\mathbb{P}(M = m) = \begin{cases} \frac{w_b \text{NB}(m; r_b, p_b)}{\sum_{j=1}^T \text{NB}(j; r_b, p_b)}, & 1 \leq m \leq T, \\ w_t p_t (1 - p_t)^{m-T-1}, & m > T. \end{cases}$$

A.19 Fitted two-regime parameters

$$T = 62, \quad r_b = 0.480, \quad p_b = 0.021, \quad p_t = 0.056, \quad w_b = 0.972, \quad w_t = 0.028.$$

A.20 Subtype HL asymptotic

$$\pi_r(X) \sim \frac{1}{|S_W(k)|} C_k \text{Li}_2(X).$$